



دانشگاه صنعتی امیر کبیر
(پلی تکنیک تهران)

Electrical and Electronic Circuits

chapter 8. Sinusoidal Steady State Analysis

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عظیم فرقدان 

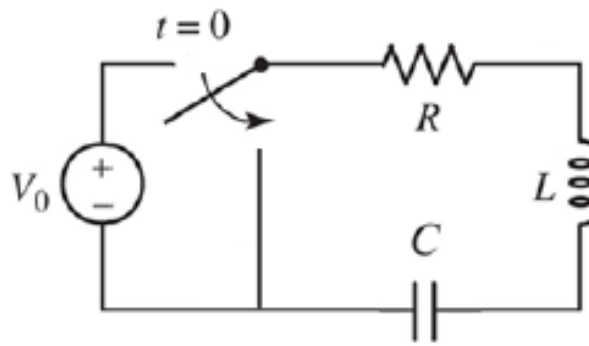
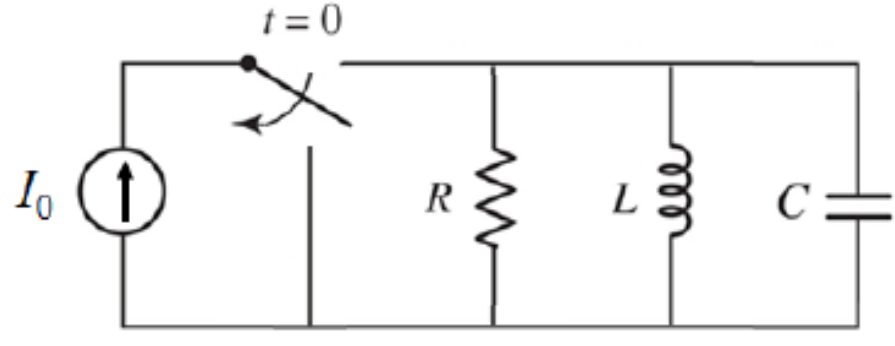
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Objectives of the Lecture

- Introduction: Sinusoidal Waves and Complex Numbers
- Force Response to Sinusoidal Input
- The Concept of the Phasor

The RL and RC Circuit

If we know that the only independent sources in our circuit are **sinusoidal**,
and we know that all transients are gone (steps, switches, pulses)...

 $\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$	 $\alpha = \frac{1}{RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}$	
NOT NECESSARY		
$\alpha > \omega_0$ OVERDAMPED $x(t) = X_1 e^{c_1 t} + X_2 e^{c_2 t} + X_3$	$\alpha = \omega_0$ CRITICALLY DAMPED $x(t) = e^{-\alpha t} (X_1 t + X_2) + X_3$	$\alpha < \omega_0$ UNDERDAMPED $x(t) = e^{-\alpha t} \left[X_1 \cos(\omega_d t) + X_2 \sin(\omega_d t) \right] + X_3$

NOT NECESSARY

...we may instead solve the circuit **algebraically** (e.g. nodal, mesh)
without determining initial conditions, final conditions, etc.

Sinusoidal Waves

Amplitude V_m

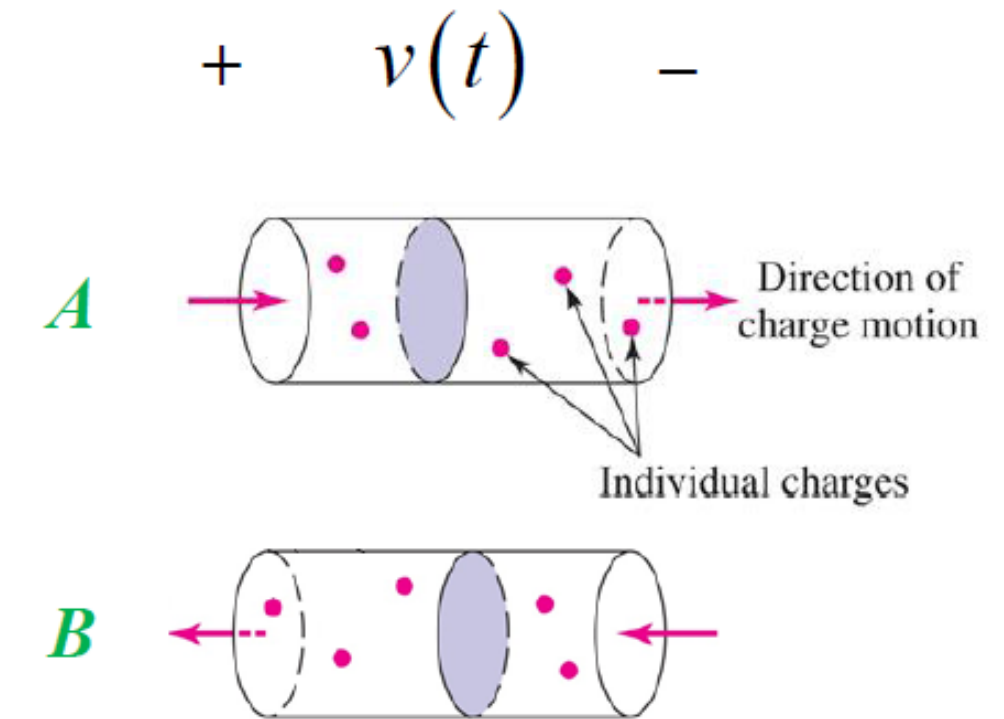
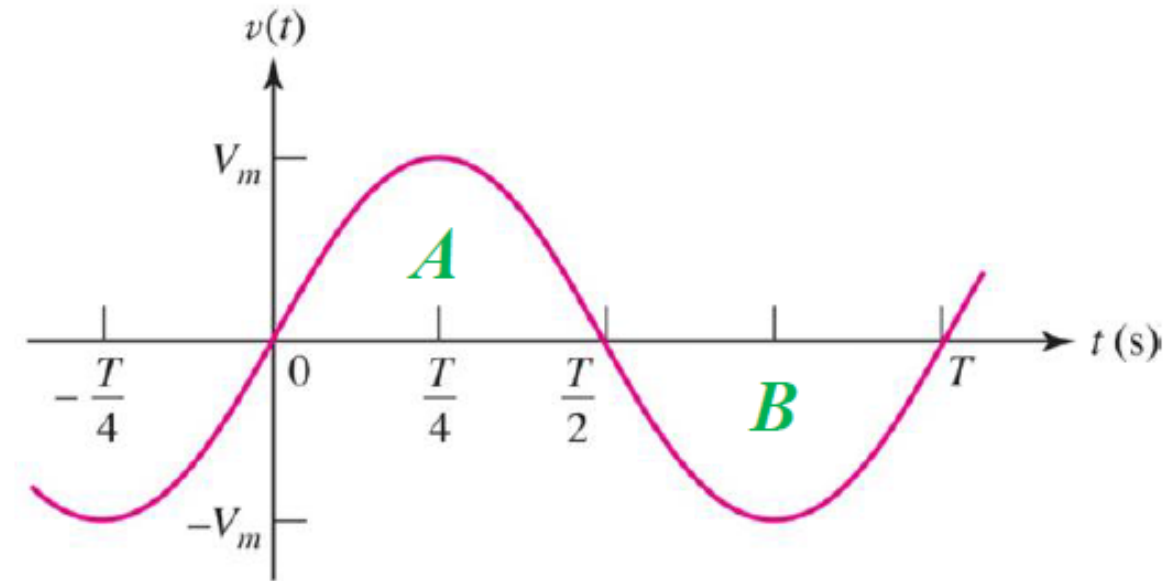
Argument ωt

Angular Frequency ω

Period T

Frequency f

$$v(t) = V_m \cdot \sin(\omega t + \phi_0) , \quad \phi_0 = 0$$



V_m = amplitude (in Volts), ϕ_0 = phase (in radians)

ω = frequency (in radians/second)

T = period (in seconds)

f = frequency (in cycles/second) = $1/T = \omega / 2\pi$

$$V_m \cdot \sin(\omega t + \phi_0) = V_m \cdot \cos(\omega t + \phi_0 - \pi/2)$$

Sinusoidal Waves

$$v(t) = V_m \sin(\omega t + \phi) \quad T = \frac{2\pi}{\omega} \quad f = \frac{1}{T}$$

where

V_m is the amplitude of the sinusoid

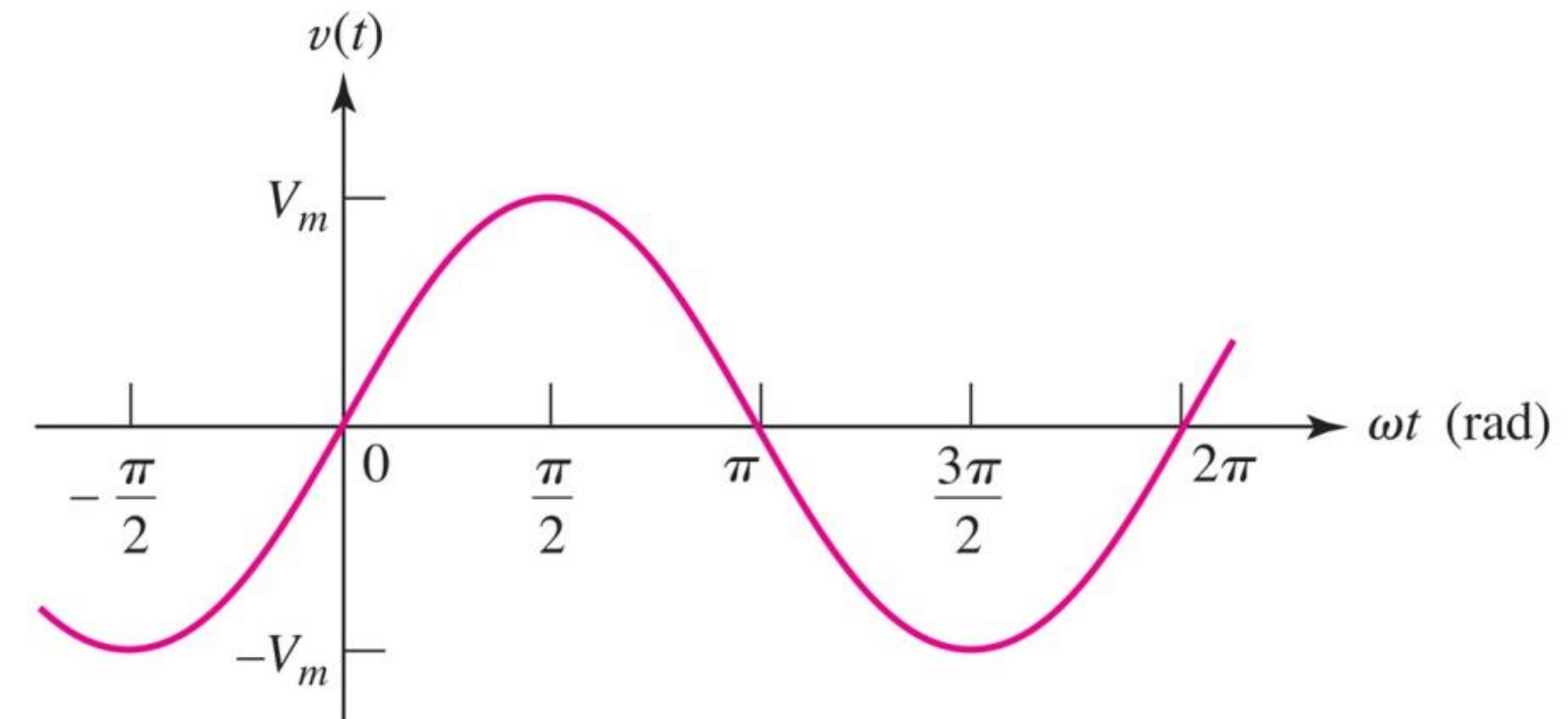
ω is the angular frequency in radians/s

ϕ is the phase angle in degrees

$\omega t + \phi$ is the argument of the sinusoid

T is the period of a sinusoid in seconds

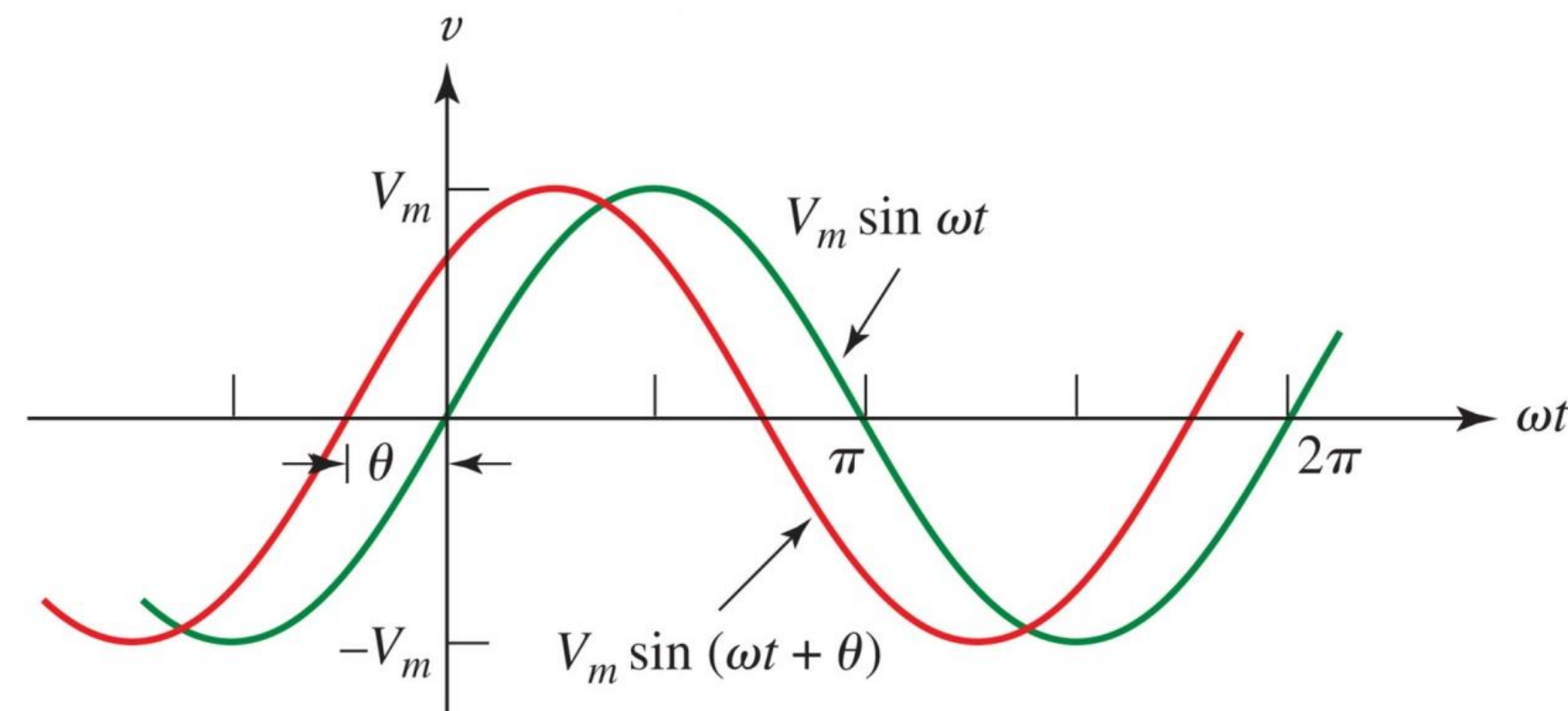
f is the frequency with units of Hz (cycles per second)



Phase of the Sinusoidal Wave

In the more general case, a sinusoidal wave includes a phase θ .

$$v(t) = V_m \sin(\omega t + \theta)$$



We say that the new wave **leads** the original wave by a phase of θ .

We say that the original wave **lags** behind the new wave by a phase of θ .

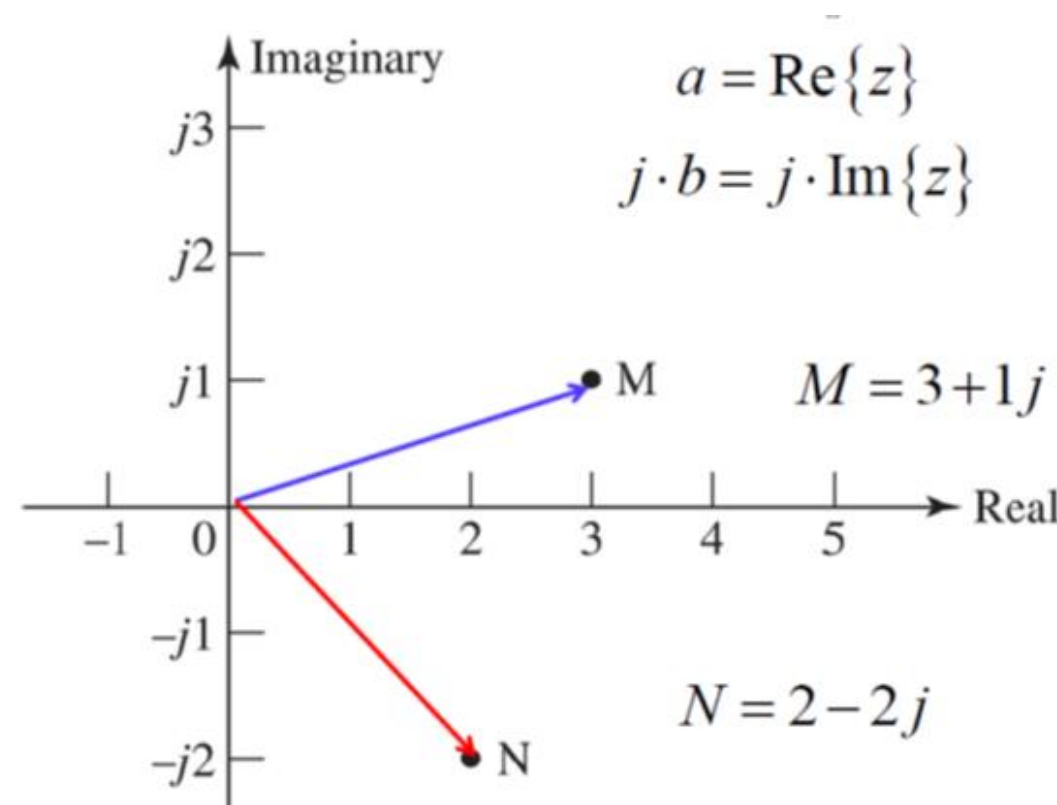
Complex numbers

- A complex number z can be written in rectangular form as

$$\diamond a + bj \leftrightarrow Ae^{j\theta}$$

$$\diamond A = \sqrt{a^2 + b^2}, \quad \theta = \tan^{-1} \frac{b}{a}$$

$$\diamond a = A \cos \theta, \quad b = A \sin \theta$$



- The complex number z can also be written in polar or exponential form as $z = A \angle \theta = Ae^{j\theta}$

$$\checkmark 2 \angle 45^\circ = 2e^{j45} = 2 \cos 45 + j2 \sin 45 = \sqrt{2} + \sqrt{2}j$$

$$\checkmark \frac{1}{1+2j} = \frac{1 \angle 0}{\sqrt{5} \angle \tan^{-1} 2} = \frac{1}{\sqrt{5}} \angle -\tan^{-1} 2$$

✓ Subtraction with Rectangular Coordinates

If $Z = X + Y$ and $X = a + jb$ and $Y = c + jd$

$$Z = (a - c) + j(b - d) \rightarrow \text{Re}(Z) = (a - c) \text{ and } \text{Im}(Z) = (b - d)$$

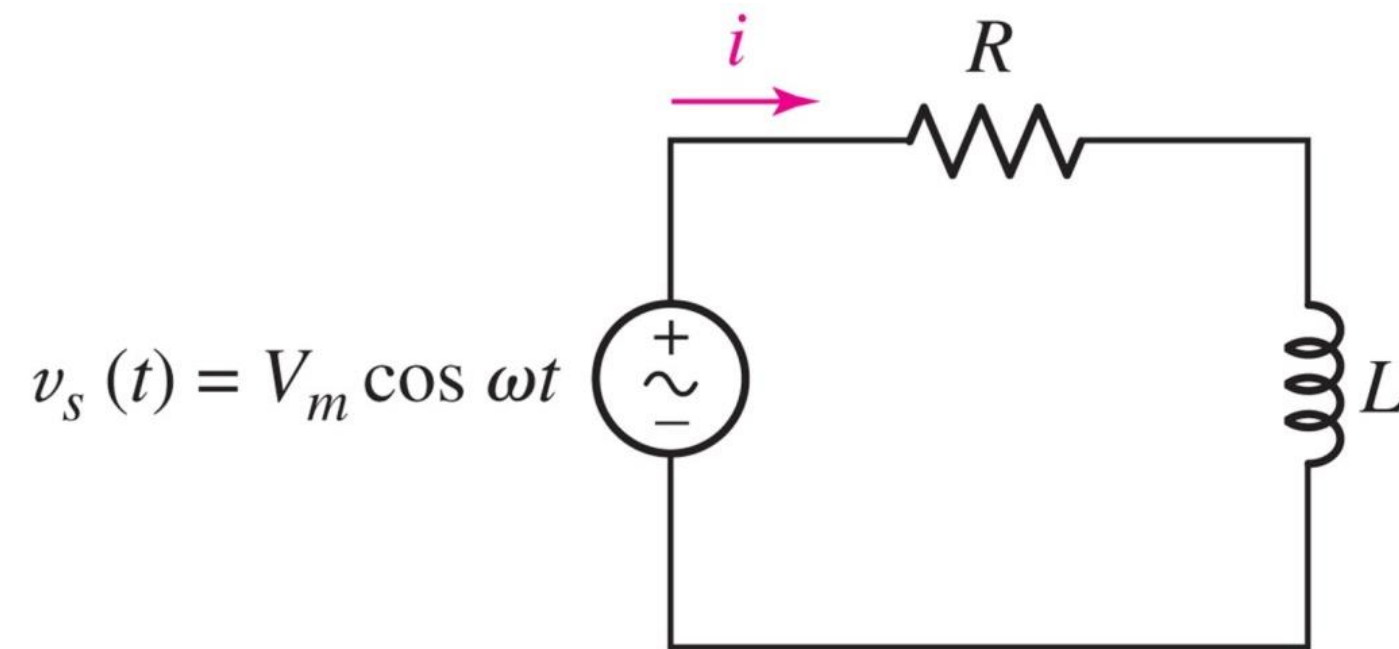
✓ Multiplication and Division

If $V = v\angle f$ and $I = i\angle q$

$$P = VI = vi \angle(f + q) \text{ and } Z = V/I = v/i \angle(f - q)$$

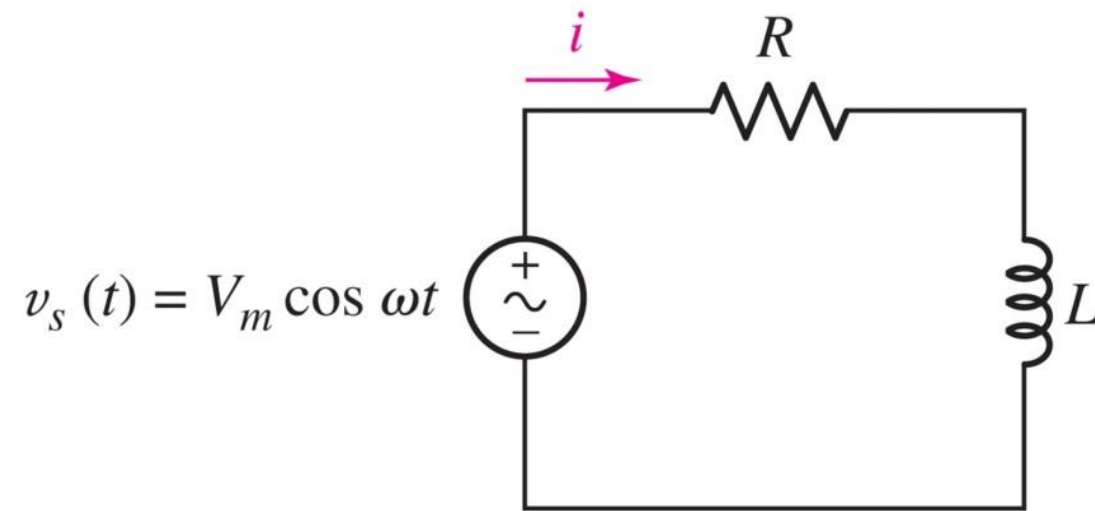
Forced Response to Sinusoidal Input

- In many applications, when the input is sinusoidal, the transient (natural) response is not of interest; instead, we are primarily concerned with finding the steady-state (forced) response.



- Here, we seek a method to **simplify obtaining this response**.

- ✓ We apply KVL (Kirchhoff's Voltage Law):



$$L \frac{di}{dt} + Ri = V_m \cos(\omega t)$$

- ✓ The forced response is of the same form as the input.

$$i(t) = I_1 \cos \omega t + I_2 \sin \omega t$$

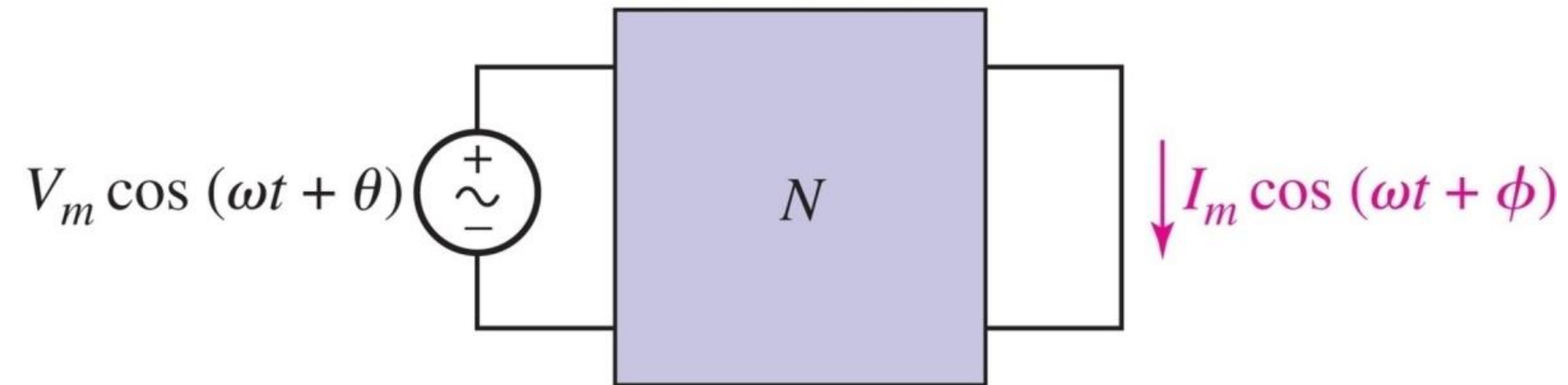
- ✓ By substituting into the differential equation, the coefficients are determined.

$$i(t) = \frac{RV_m}{R^2 + \omega^2 L^2} \cos \omega t + \frac{\omega L V_m}{R^2 + \omega^2 L^2} \sin \omega t$$

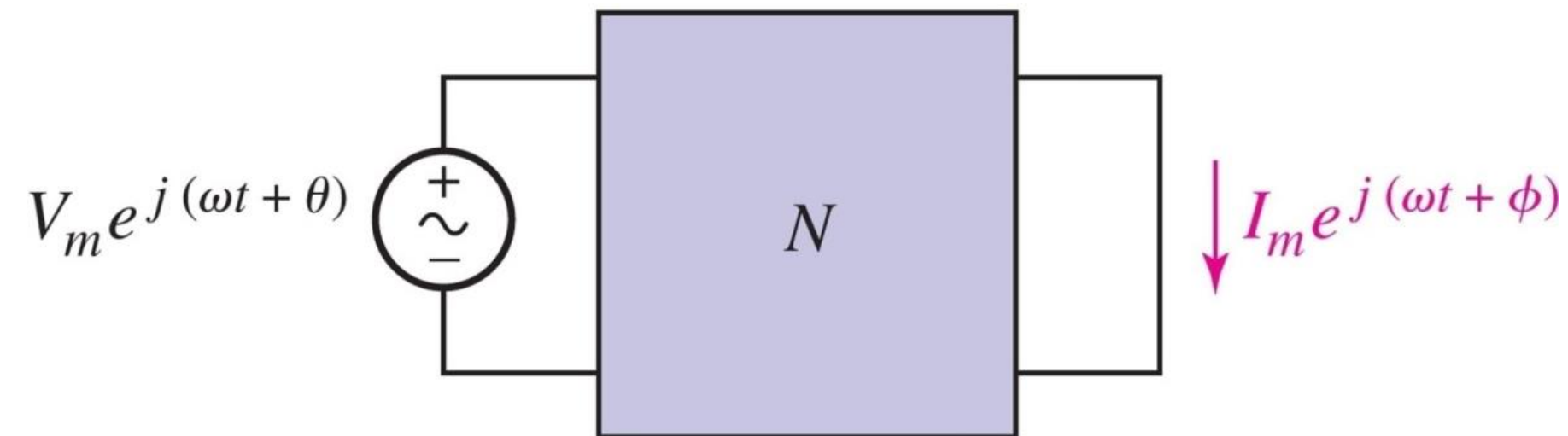
Conclusion?

- ✓ The forced response to a sinusoidal input can be **obtained as before** by solving the **differential equation**.
- ✓ Is there a way to **avoid differential equations** and rely solely on **algebraic calculations** to find the forced sinusoidal response?
- ✓ Yes, by using the concept of the **phasor**!

- Main problem: finding the steady-state response of circuit N to a sinusoidal input.



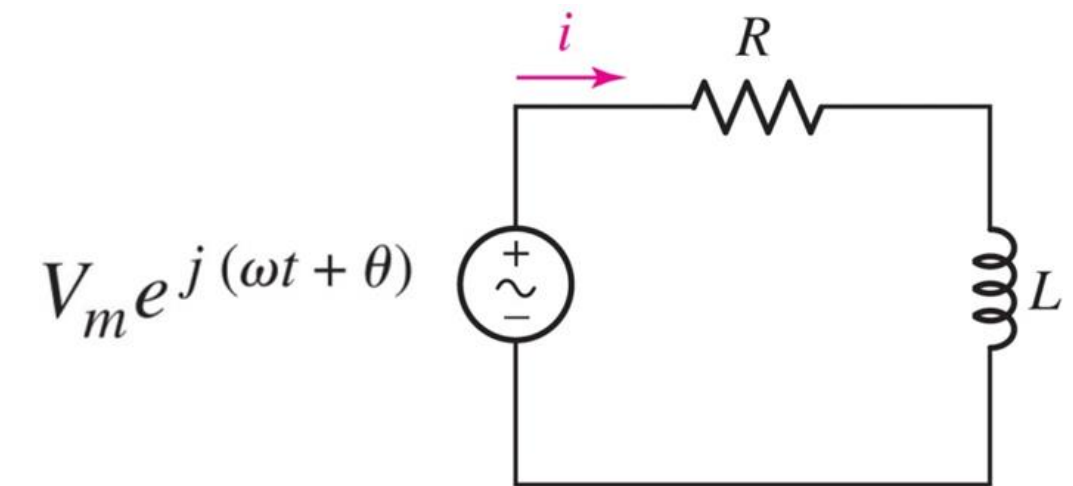
- Alternative approach: let us determine the response of the circuit to the following exponential input:



Forced Response to Complex Exponential Input

1. By applying KVL, we have:

$$L \frac{di}{dt} + Ri = v_s$$



2. The forced response is of the same form as the output.

$$i(t) = I_m e^{j(\omega t + \phi)}$$

3. Substituting into the differential equation:

$$\triangleright j\omega L I_m e^{j(\omega t + \phi)} + R I_m e^{j(\omega t + \phi)} = V_m e^{j(\omega t + \theta)}$$

$$\triangleright j\omega L I_m e^{j\phi} + R I_m e^{j\phi} = V_m e^{j\theta}$$

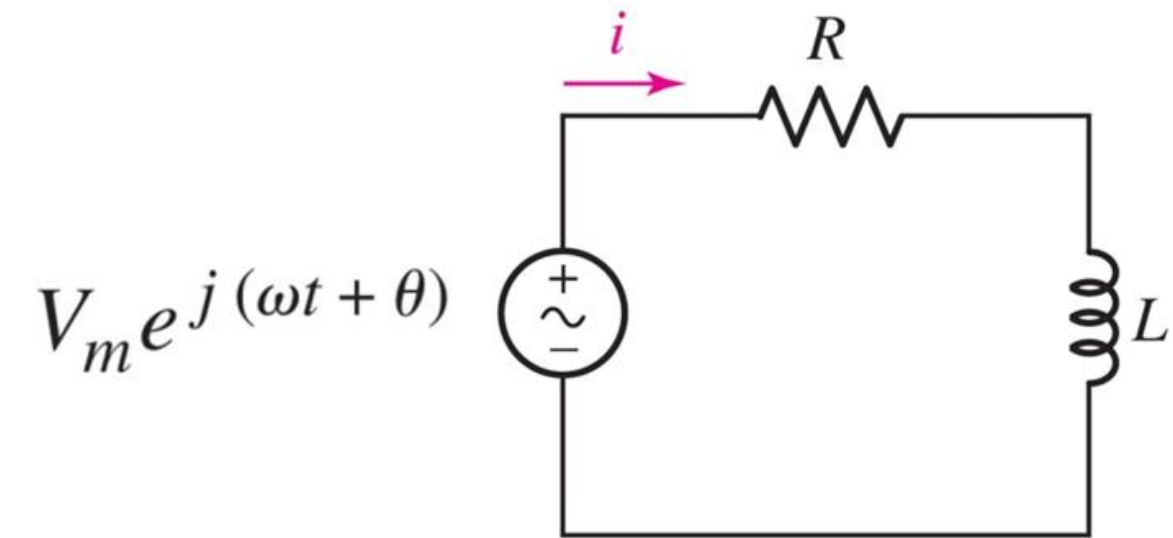
$$\triangleright I_m e^{j\phi} = \frac{V_m e^{j\theta}}{R + j\omega L} \rightarrow I_m = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}}, \quad \phi = \theta - \tan^{-1} \frac{\omega L}{R}$$

$$\triangleright i(t) = \text{Re}[I_m e^{j(\omega t + \phi)}] = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \theta - \tan^{-1} \frac{\omega L}{R})$$

The concept of phasor

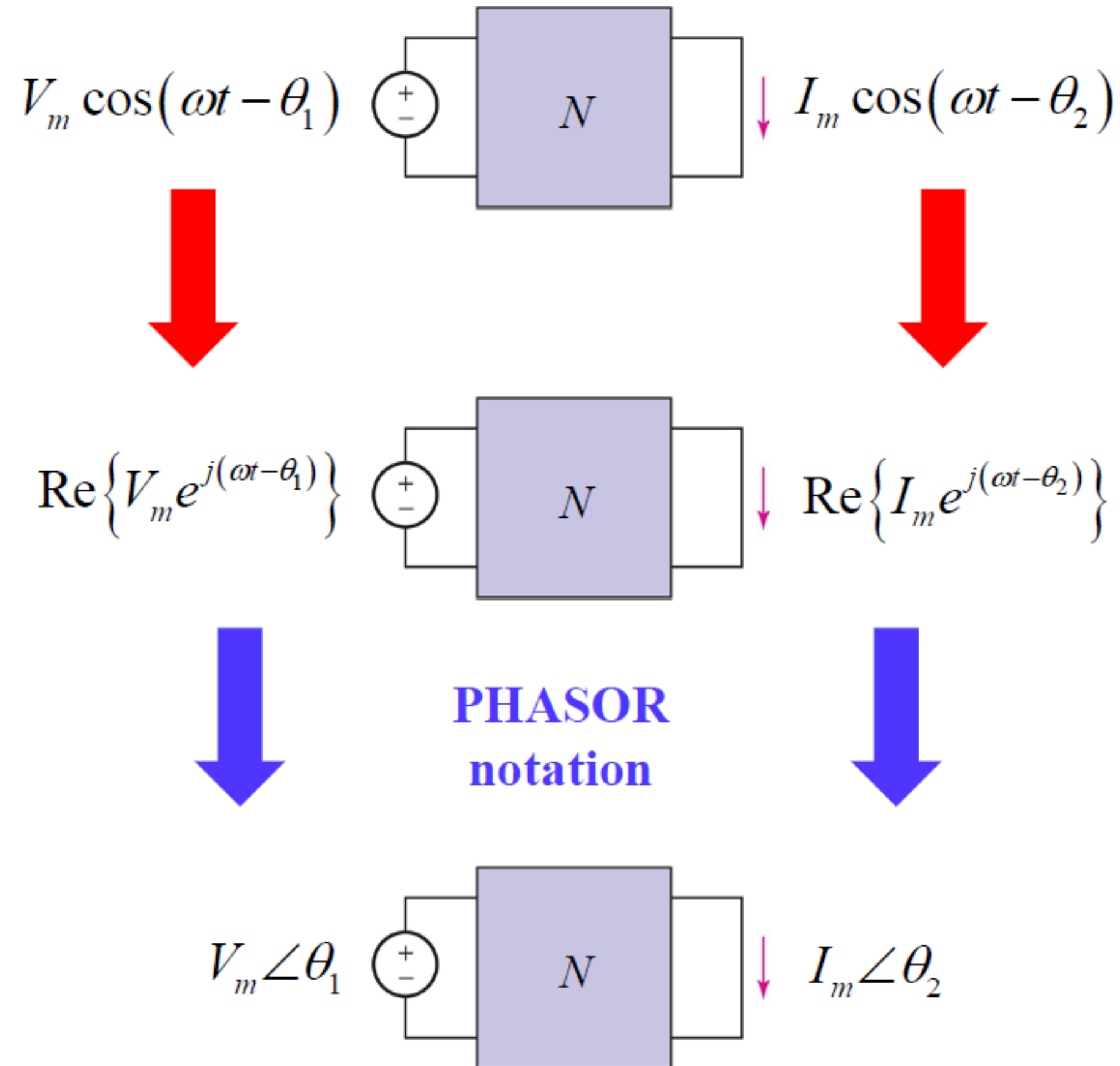
- ✓ In the sample circuit shown, we arrive at the following relationship:

$$I_m e^{j\phi} = \frac{V_m e^{j\theta}}{R + j\omega L}$$



- ✓ A complex number with a magnitude of V_m and an angle θ , represented as $V_m e^{j\theta}$, is referred to as a **phasor**.
- ✓ Similarly, the phasor for the current is $I_m e^{j\phi}$.
- ✓ The **relationship between the phasors** in the circuit is an **algebraic relationship**, not a differential one!

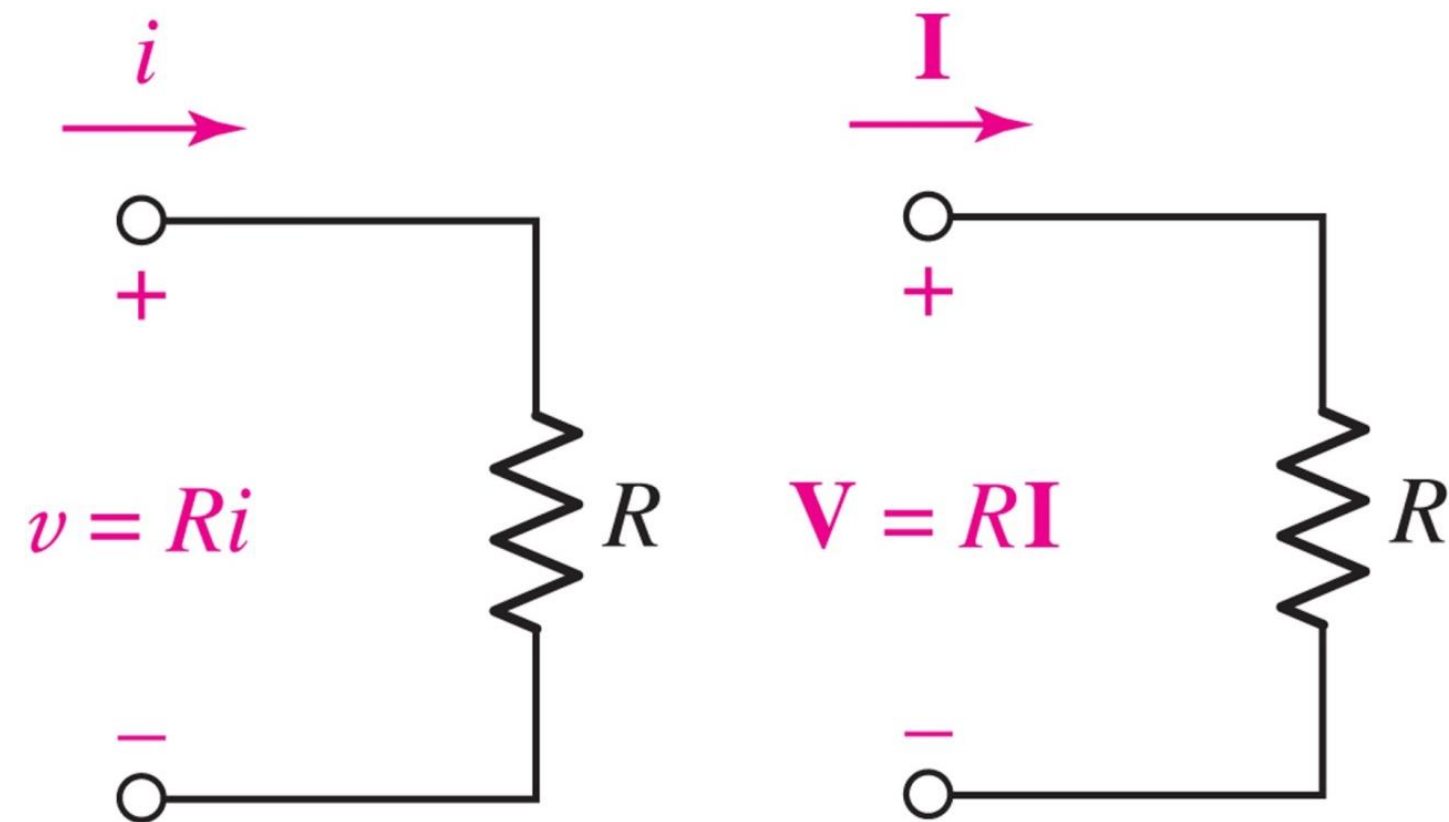
Sinusoidal vs. Complex Representation



The phasor of the resistor

The voltage phasor and current phasor of a resistor also follow Ohm's Law.

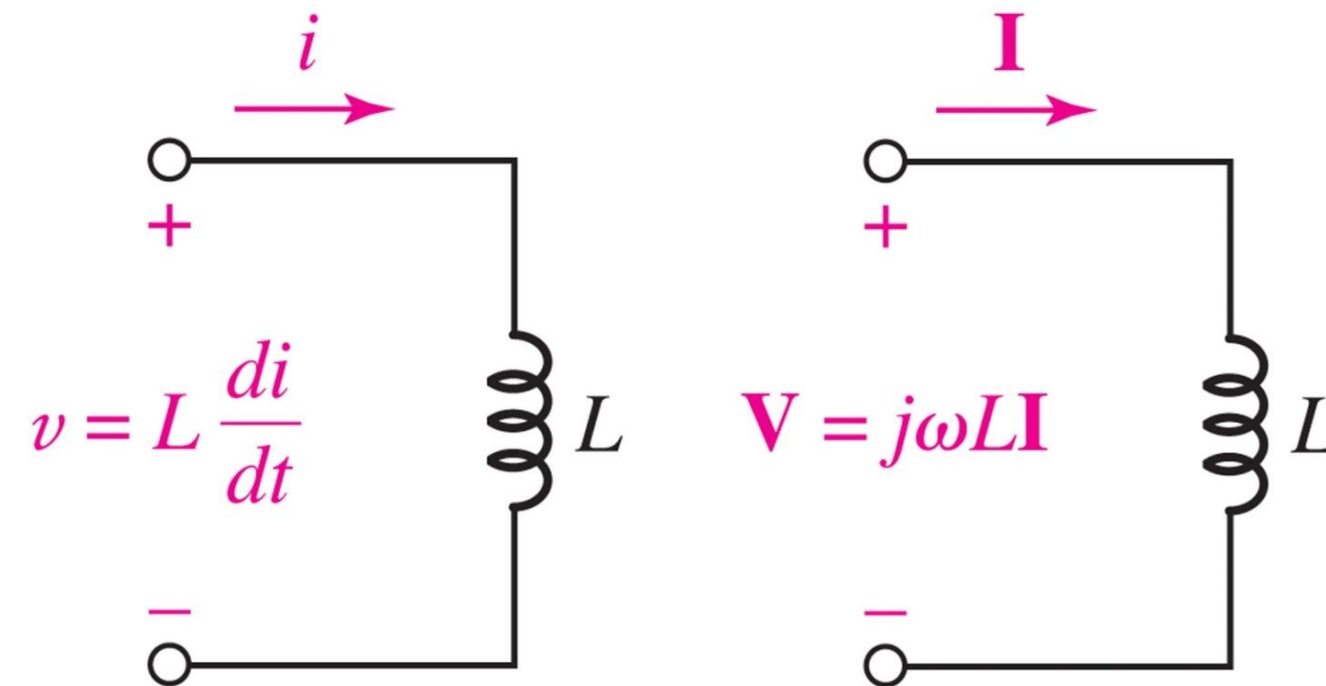
Thus, the phasor of the resistor is simply R .



The phasor of the inductor

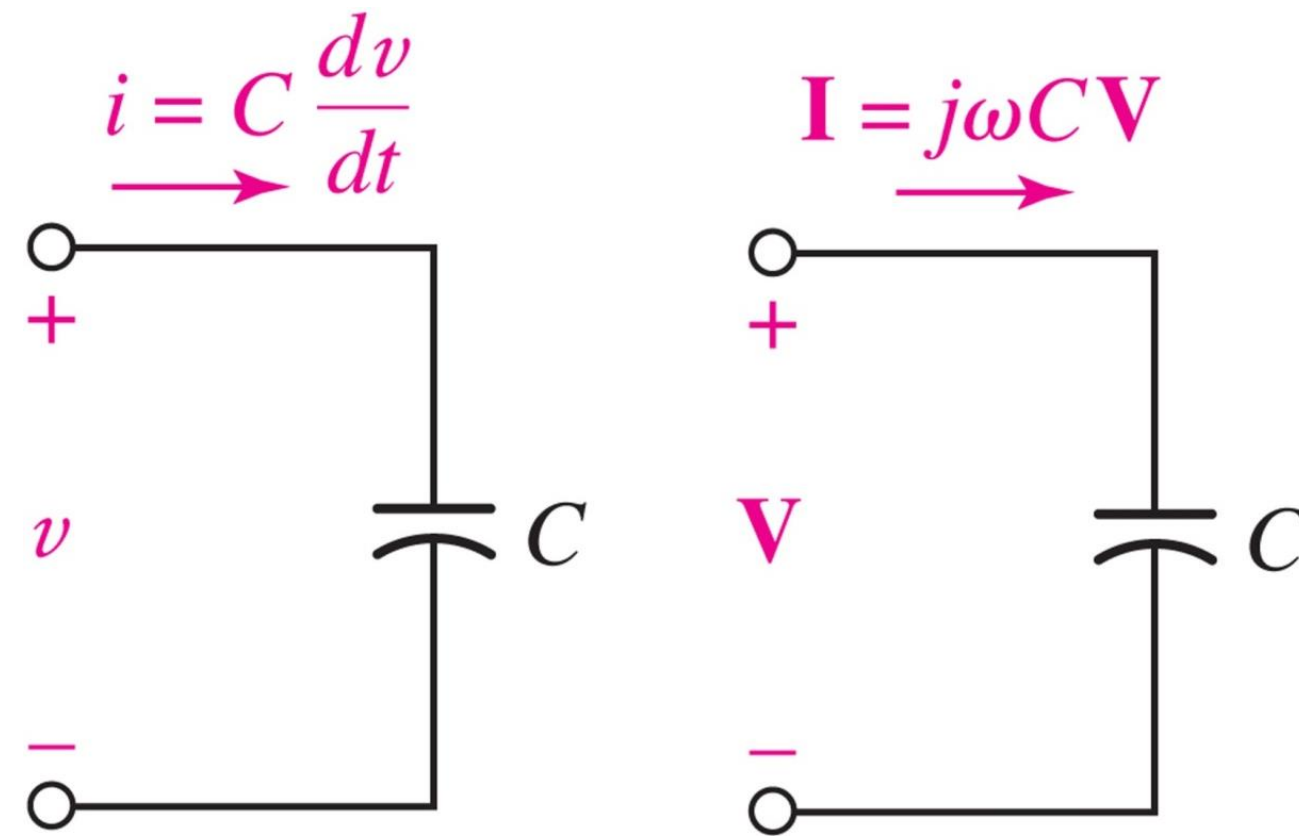
The differential relationship between the current and voltage of an inductor in the time domain transforms into an algebraic relationship in the phasor domain.

The phasor of the inductor is $j\omega L$, and its voltage and current phasors satisfy Ohm's law.

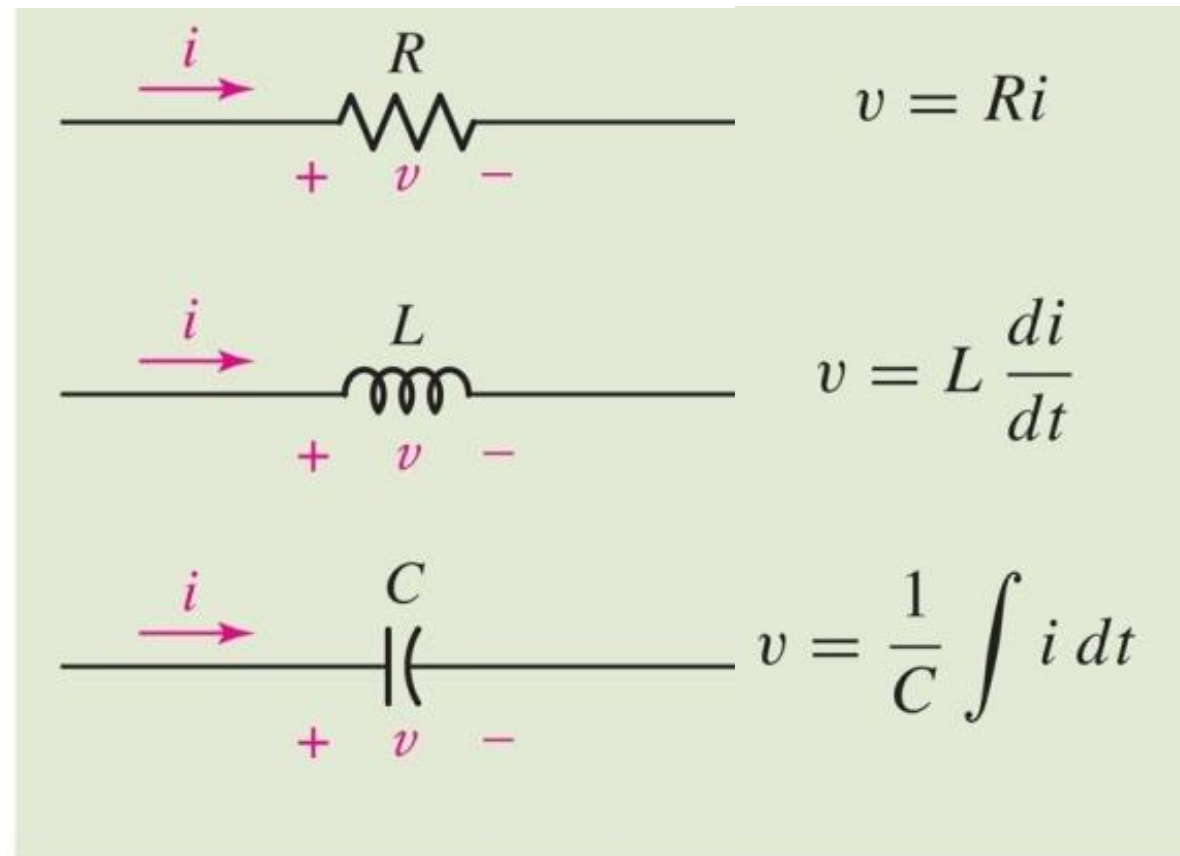


The phasor of the capacitor

The phasor of the capacitor is $\frac{1}{j\omega C}$, and the relationship between its voltage and current in the phasor domain, like that of the inductor and resistor, follows Ohm's law.

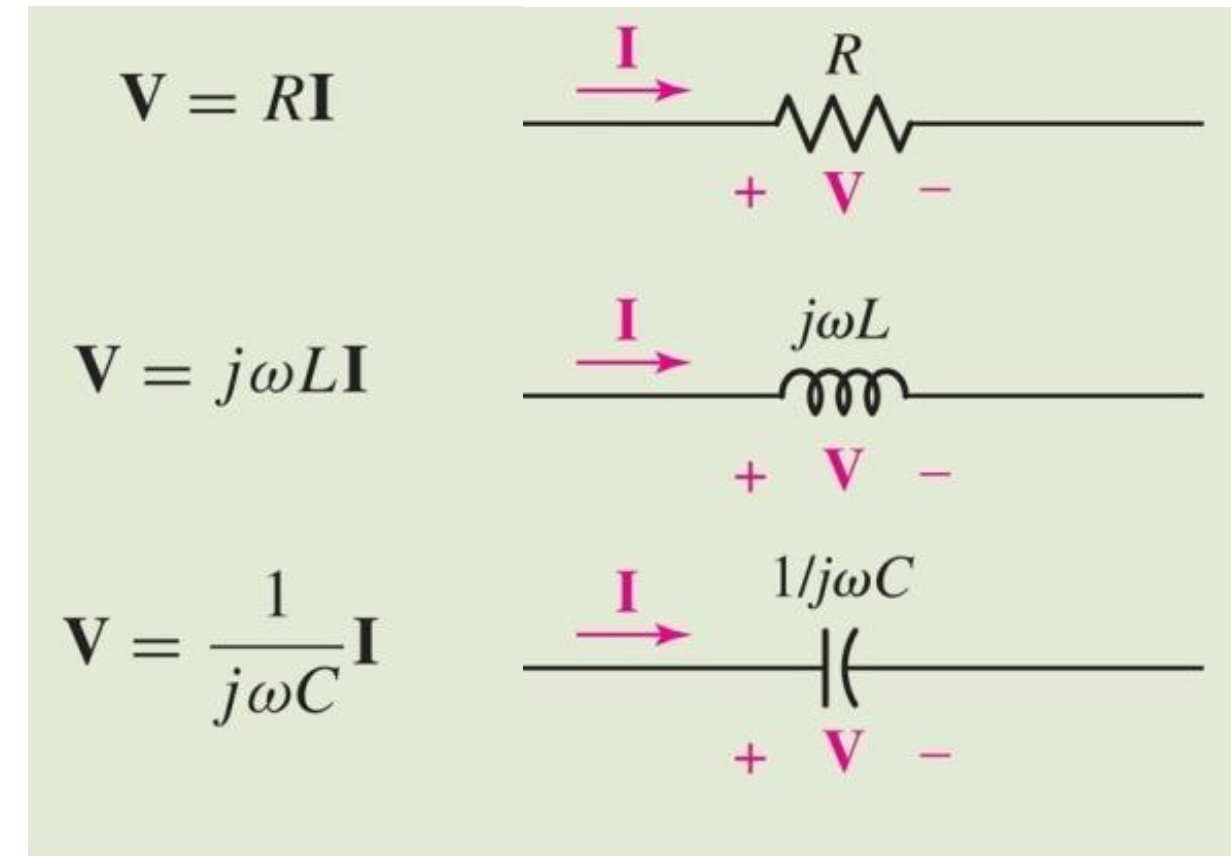


Time Domain



Differential calculations with
real numbers

Phasor Domain



Algebraic calculations with
complex numbers

- The KVL relationship also holds for voltage phasors in a loop.

$$\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_N = 0$$

- Similarly, the KCL relationship holds for current phasors at a node.

$$\mathbf{I}_1 + \mathbf{I}_2 + \cdots + \mathbf{I}_N = 0$$

Impedance

- ✓ The ratio of the voltage phasor to the current phasor is called impedance.

$$Z_R = R \quad Z_L = j\omega L \quad Z_C = 1/j\omega C$$

- ✓ Impedance is equivalent to resistance in the phasor domain.
- ✓ Impedance is a complex number and is measured in ohms.
- ✓ The **real part** of impedance is called **resistance**, and **the imaginary** part is called **reactance**.
Series and parallel impedances can be combined in the same way as resistances.

- The inverse of impedance is called admittance.

$$Y_R = 1/R \quad Y_L = 1/j\omega L \quad Y_C = j\omega C$$

- Admittance is equivalent to conductance.
- Admittance is a complex number, and its unit is Siemens.
- The **real** part of admittance is called **conductance**, and the **imaginary part** is called **susceptance**.

Summary of the Phasor Method

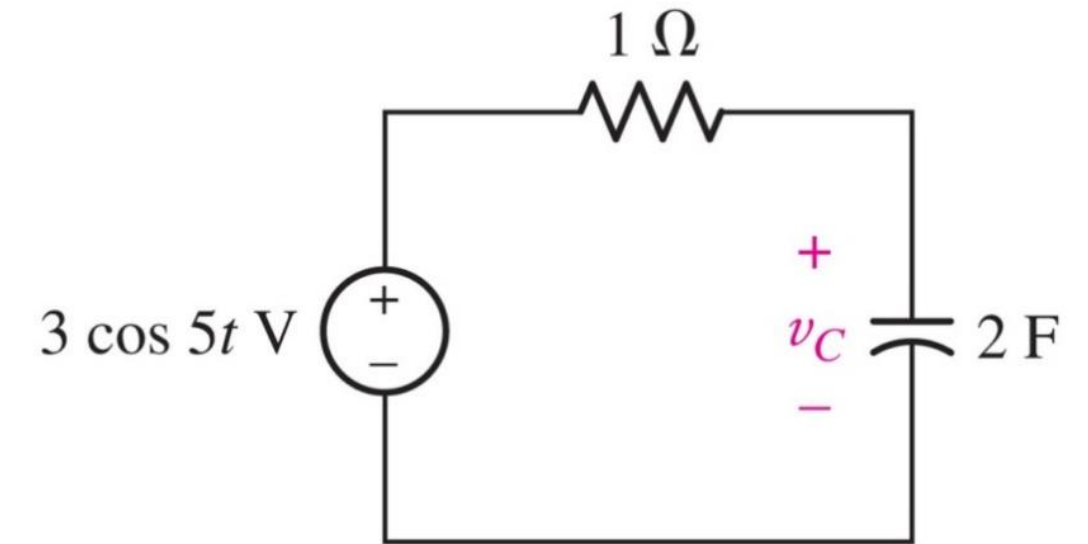
1. Determine the impedance of all elements based on the source frequency.

$$Z_R = R \quad Z_L = j\omega L \quad Z_C = 1/j\omega C$$

1. Replace the source values with their corresponding phasors.
2. Analyze the circuit in the phasor domain as if it were a resistive circuit, and find the phasors for all currents and voltages in the circuit.
3. Converting phasors back to the time domain: Multiply the desired phasor by $e^{j\omega t}$ and take the real part as the final response.

Example 1: Using Phasors

Find the steady-state response of the voltage across the capacitor.



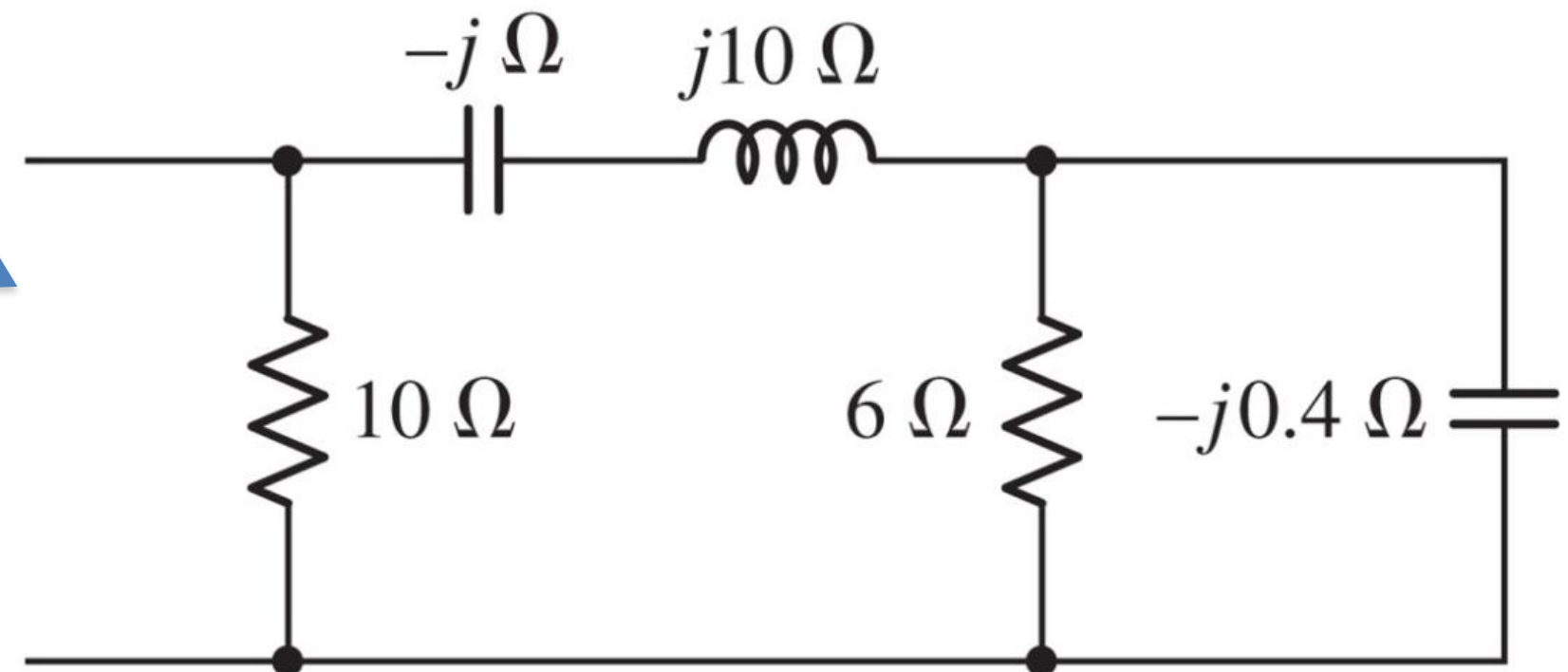
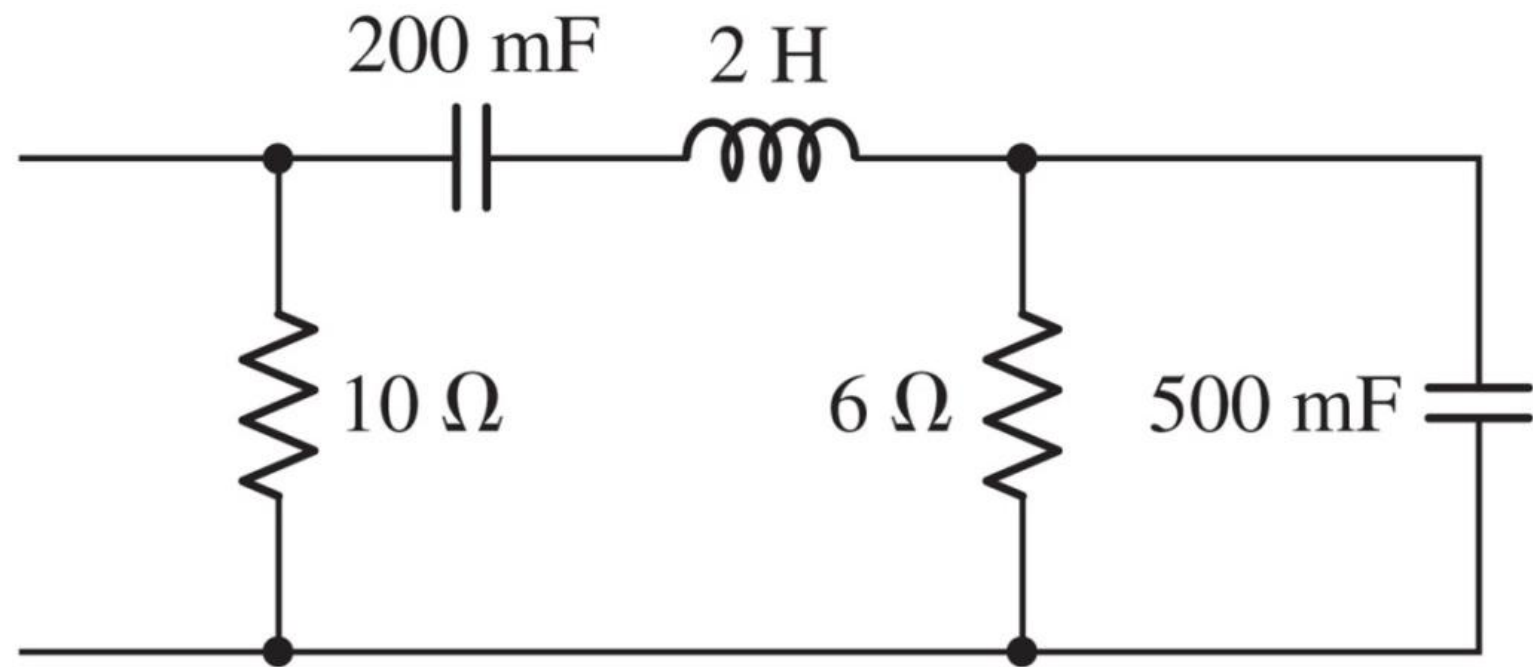
$$V_s = 3\angle 0, \quad Z_R = 1, \quad Z_C = \frac{1}{j\omega C} = \frac{1}{10j} = -0.1j$$

$$V_c = \frac{-0.1j}{1 - 0.1j} \times 3 = \frac{3}{1 + 10j} = \frac{3\angle 0}{\sqrt{101}\angle \tan^{-1} 10} = 0.298\angle -84.3$$

$$v_C(t) = \text{Re}[0.298e^{j(5t-84.3)}] = 0.298 \cos(5t - 84.3)$$

Example 2: Calculation of Equivalent Impedance

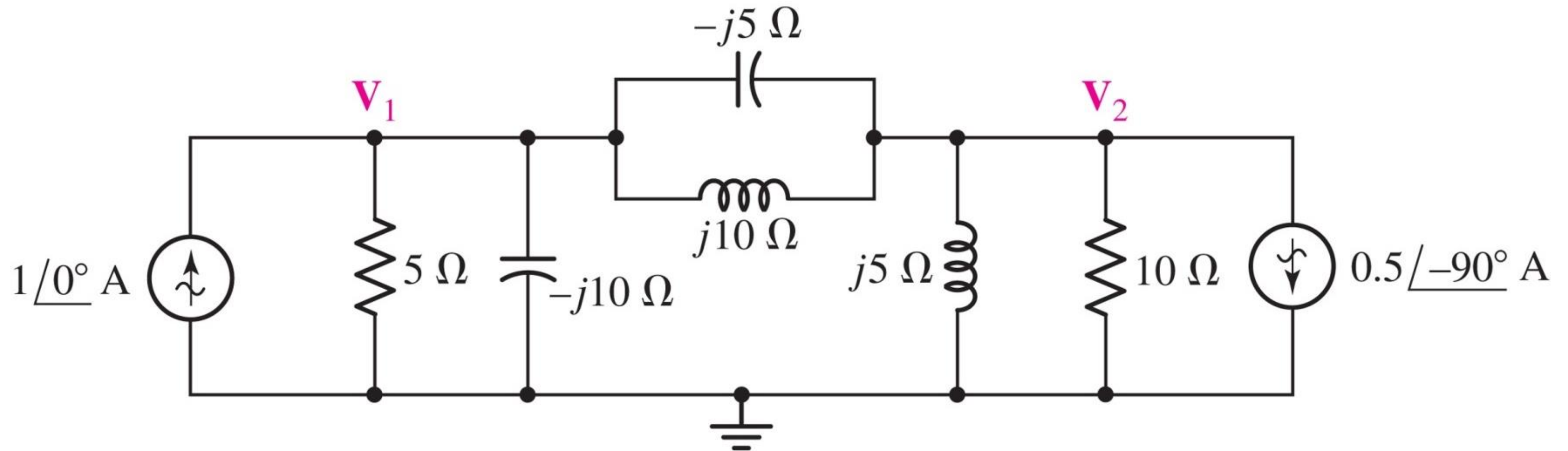
Determine the equivalent impedance of the circuit below at a frequency of 5 rad/s.



Answer: $4.255 + j4.929\ \Omega$

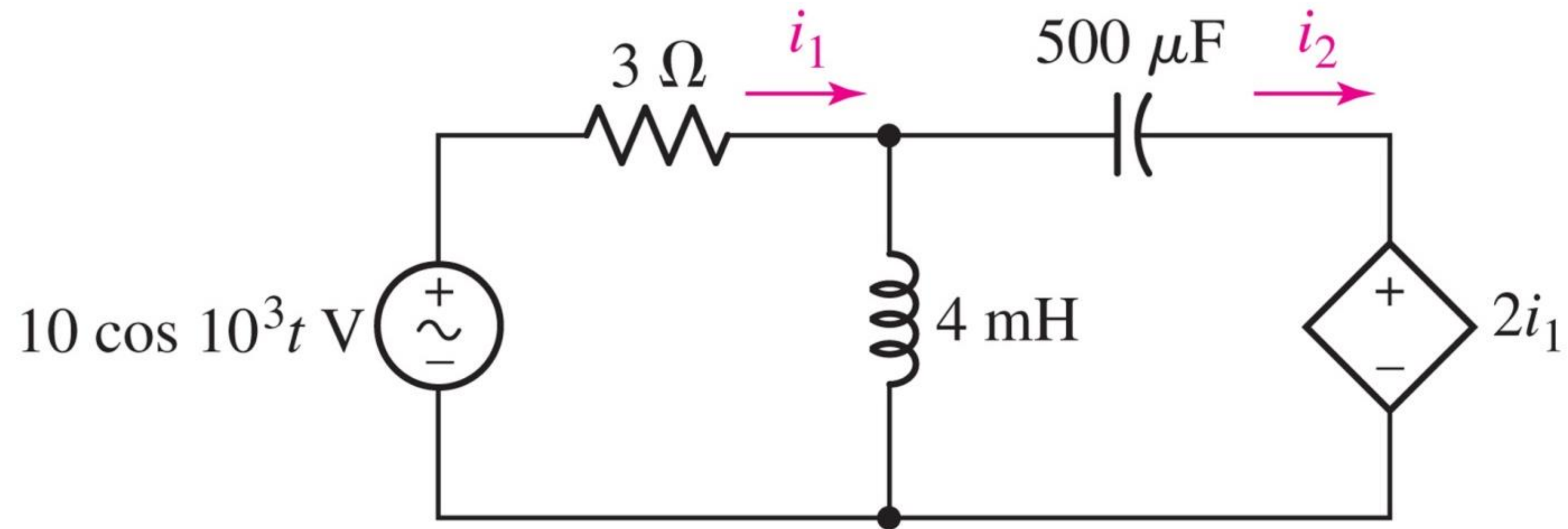
Example 3: Node Analysis

Determine the phasors for the voltages V_1 and V_2 .



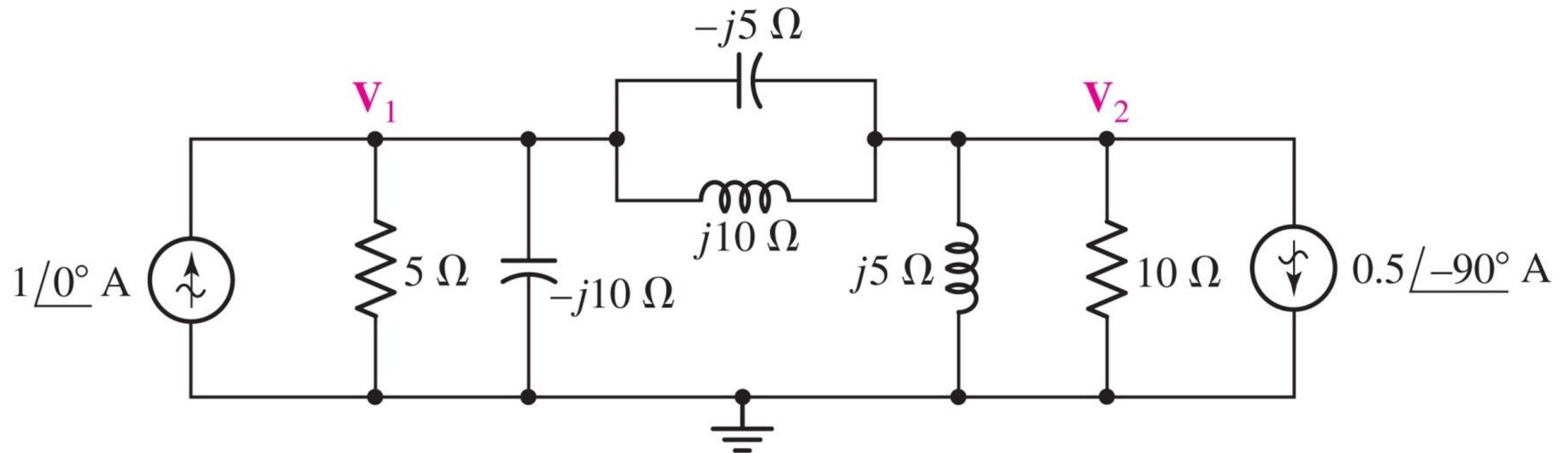
Example 4: Mesh Analysis

Determine the currents $i_1(t)$ and $i_2(t)$.



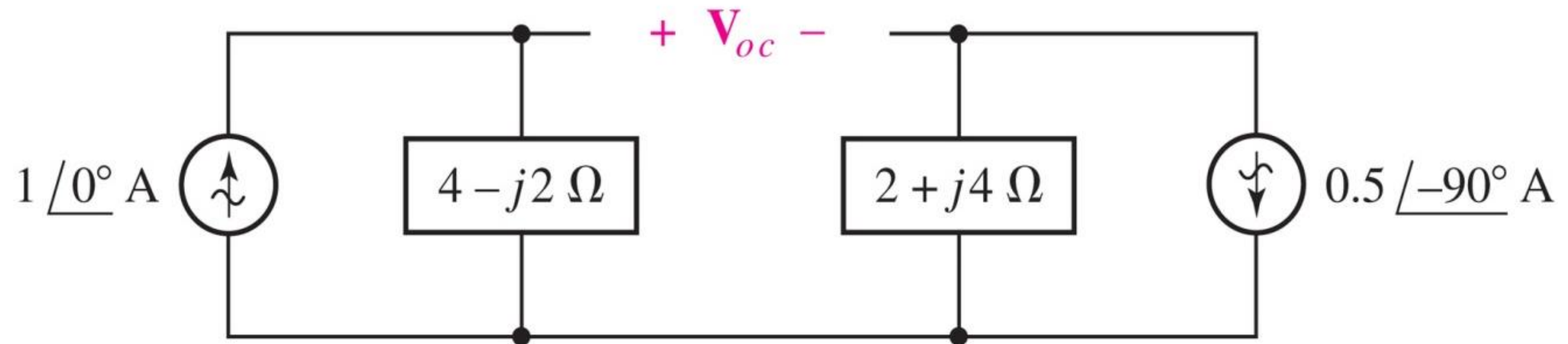
Example 5: Superposition Theorem

The principle of superposition is also valid for phasors. Use this principle to determine the voltage V_1



Example 6: Thevenin Equivalent Circuit

Theorems of Thevenin and Norton are also applicable to phasors. By calculating the Thevenin equivalent circuit of the given diagram, first determine the current flowing between nodes V1 and V2, and then find the voltage V1.



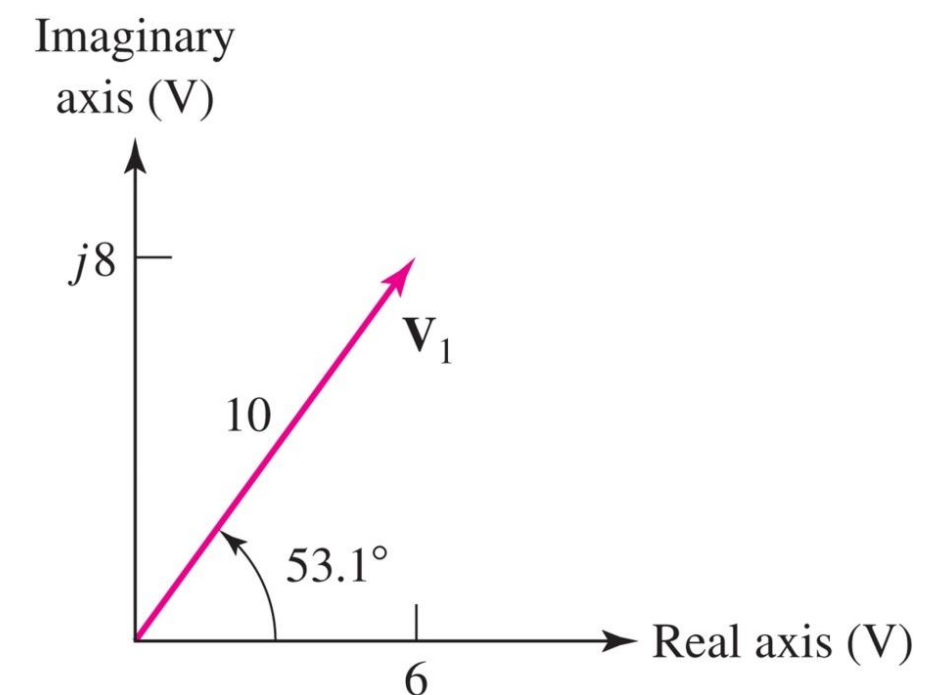
Phasor Vector Diagram

- Assume we have a phasor vector in the complex plane that rotates around the origin with an angular speed ω .
- The figure below illustrates the position of this vector at the moment $t = 0$. This represents the phasor voltage V_1 with a magnitude of 10 and an angle of 53.1 degrees.

$$V_1 = 10e^{j53.1}$$

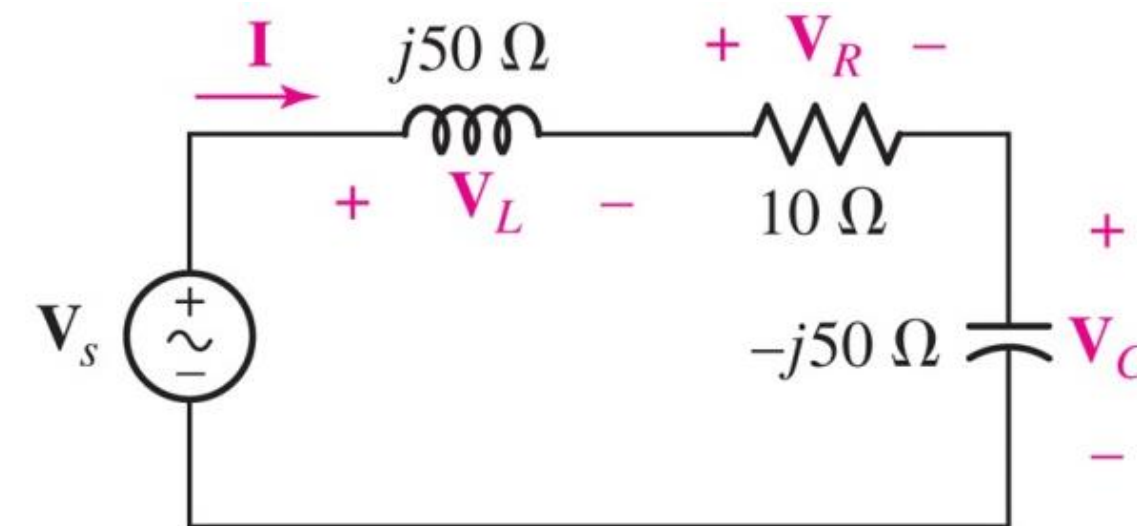
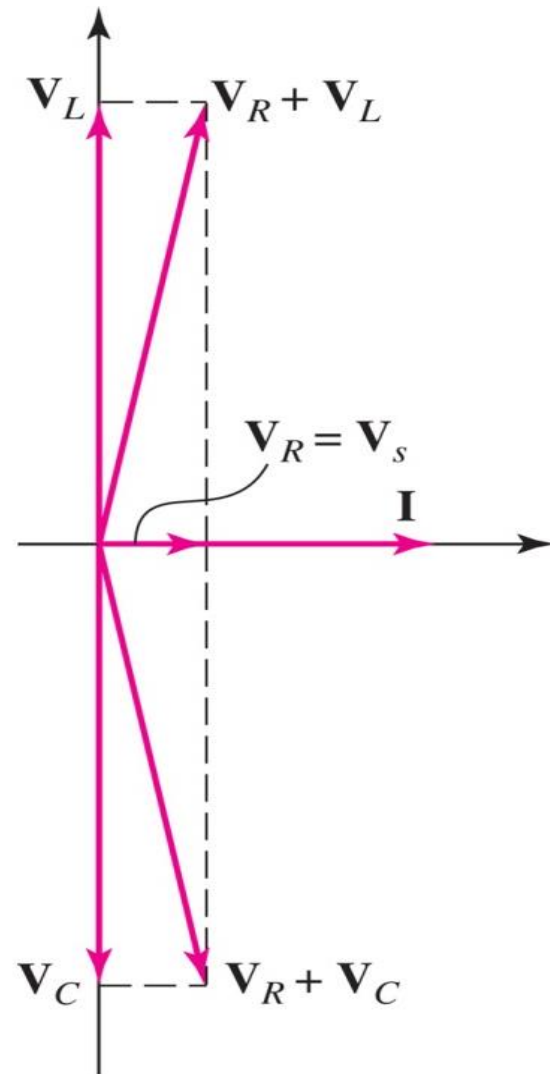
- The projection of this vector onto the real axis at any moment indicates the instantaneous voltage $v_1(t)$:

$$v_1(t) = 10 \cos(\omega t + 53.1)$$



Phasor Vector Diagram: Example 1

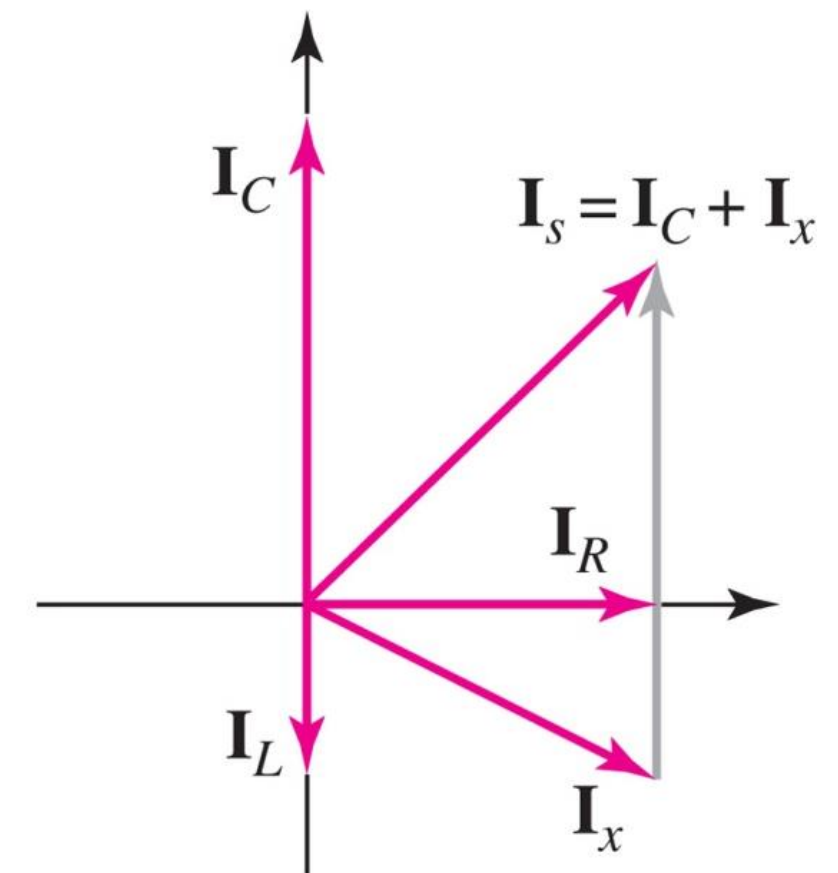
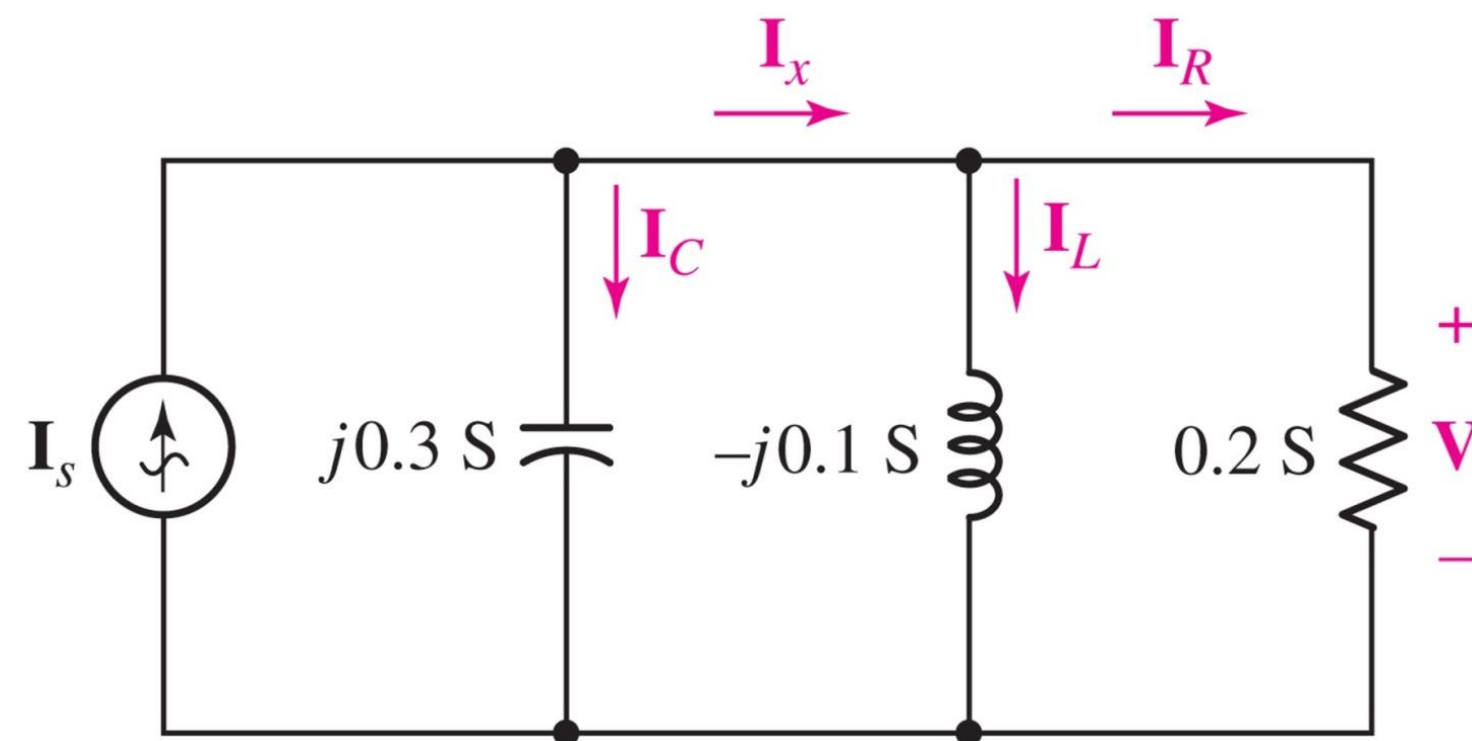
Assuming $I = 1\angle 0$



In this context, the behaviour of inductors and capacitors can be understood in terms of their phase relationships with voltage.

Phasor Vector Diagram: Example 2

Assuming $V = 1\angle 0$:



- Ohm's Law

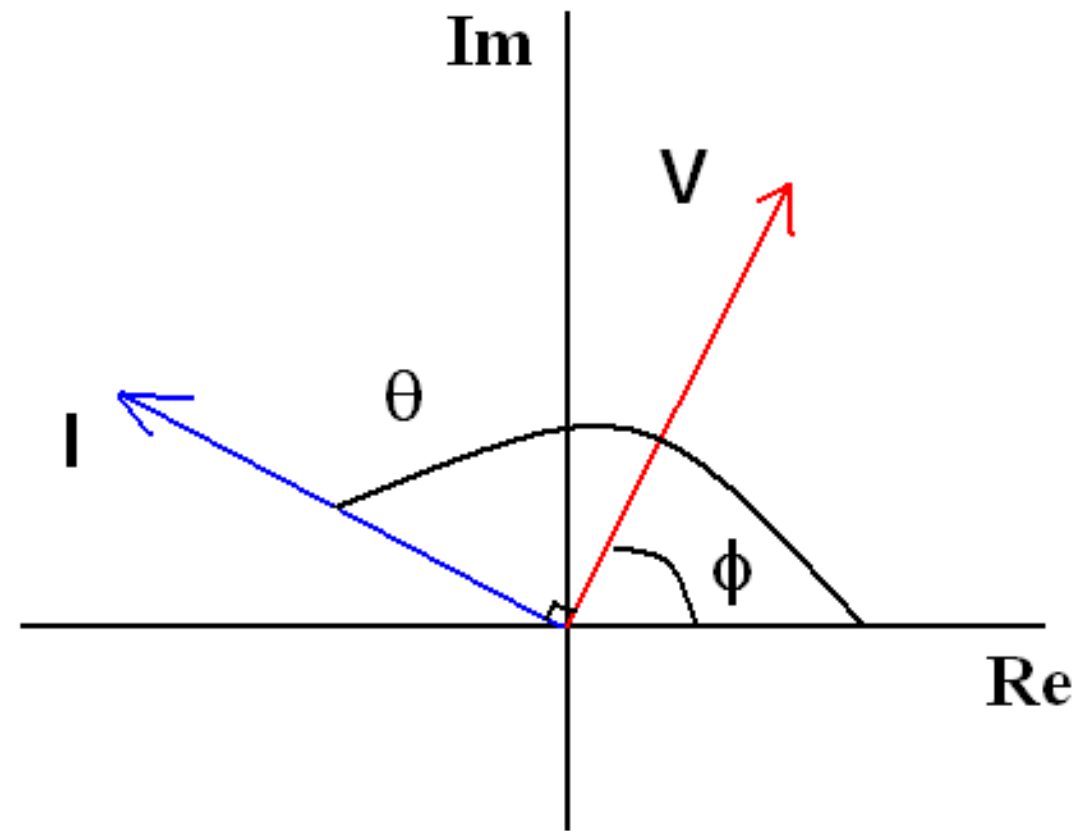
$$v(t) = Ri(t) = R I_m \cos(\omega t + \phi)$$

$$V = RI_m \quad \phi = 0 \text{ where } \phi = 0$$

- The voltage and current through a resistor are in phase as there is no change in the phase angle between them.

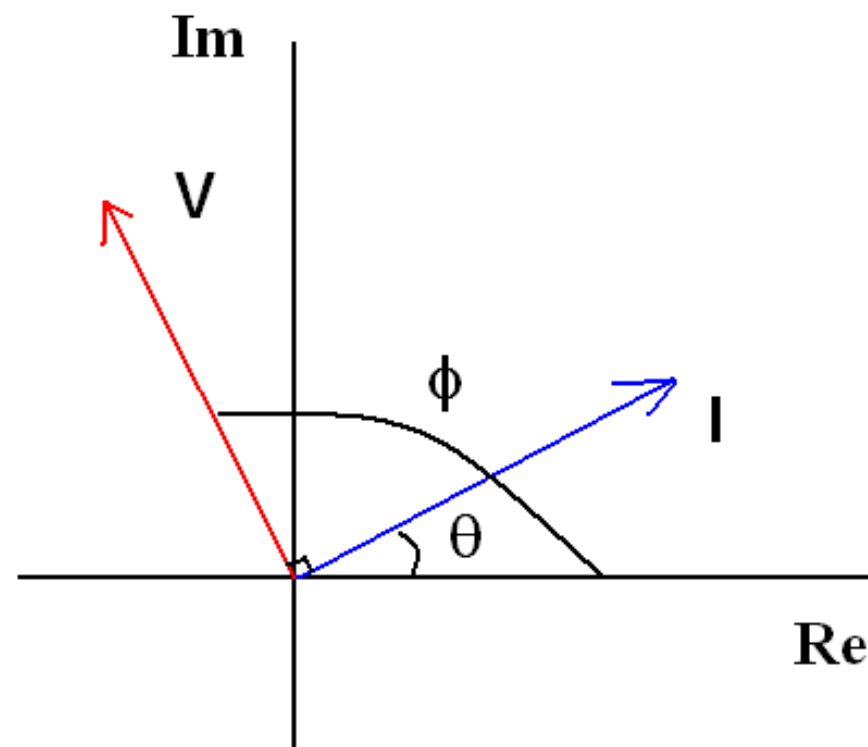
Capacitors

- 90° phase difference between the voltage and current through a capacitor.
- Current leads the voltage (or the voltage lags the current) in a capacitor.



Inductors

- 90° phase difference between the voltage and current through an inductor.
- The voltage leads the current (or the current lags the voltage).



Summary

Ohm's Law can be used to determine the ac voltages and currents in a circuit.

- ✓ Voltage leads current through an inductor.
- ✓ Current leads voltage through a capacitor.

Component		Impedance		Admittance	
Resistor	Z_R	R	$R \angle 0^\circ$	G	$G \angle 0^\circ$
Capacitor	Z_C	$-j/\omega C$	$1/\omega C \angle -90^\circ$	$j\omega C$	$\omega C \angle +90$
Inductor	Z_L	$j\omega L$	$\omega L \angle +90$	$-j/\omega L$	$1/\omega L \angle -90^\circ$

Question: What should be done if a circuit contains sources with different frequencies?

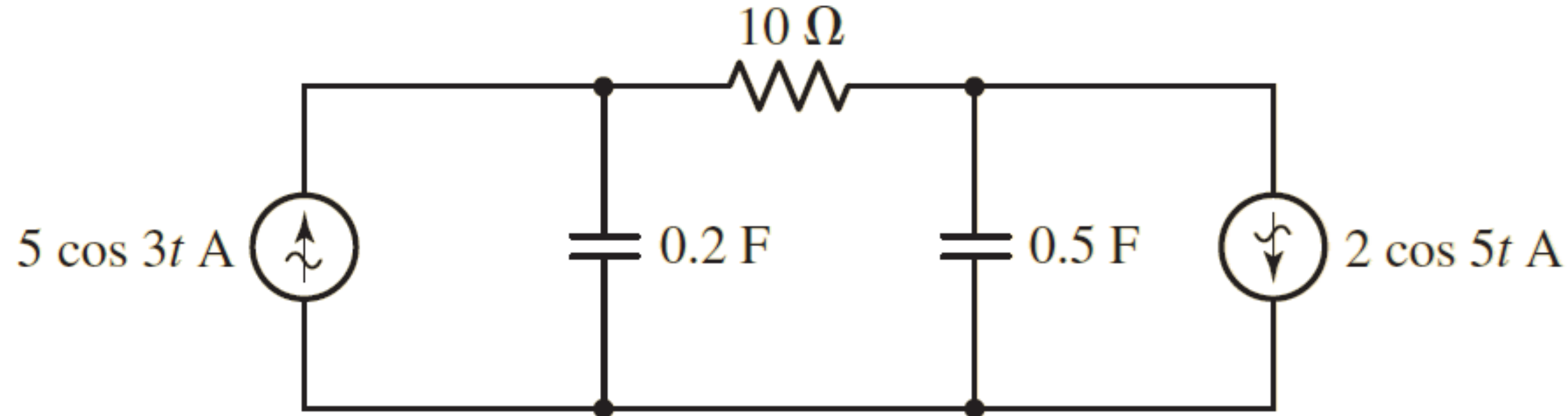
When calculating $j\omega L$ and $\frac{1}{j\omega C}$, which ω should be used?

Answer: In this case, the effect of each source must be determined separately, and the responses should be combined in the time domain (using the principle of superposition).

Important: Note that phasors corresponding to two different frequencies cannot be directly summed! They must first be transformed to the time domain and then added together.

Practice 2

Determine the power consumed by the 10-ohm resistor in the circuit shown below under steady-state conditions.





Thanks
